Heterogeneous Cuckoo Search-Based Unsupervised Band Selection for Hyperspectral Image Classification

Meng Wu, Xianfeng Ou, Youli Lu, Wujing Li, Dan Yu, Zhihao Liu, and Chengtao Ji, *Member, IEEE*

***Abstract*— Hyperspectral image (HSI) characteristics of the abundant spectral information are favored by many scholars, but the challenge is how to select relevant features from such high-dimensional data. Band selection (BS), one of the most fundamental dimensionality reduction (DR) techniques, removes redundant bands while providing a subset of bands that can preserve high information content and low noise for further HSI classification. Cuckoo search (CS) algorithm is well known for its high performance of searching relevant features but struggles to get rid of local extremes in the late iteration. Therefore, in this article, an unsupervised BS method based on the heterogeneous CS algorithm with matched filter (HCS-MF) is proposed for HSI classification, in which an optimization model is constructed based on the sensitivity of the matching filter to noise. To reduce the similarity between selected bands, a mapping method based on neighborhood band grouping (NBG) is proposed. In addition, an automatic recommendation strategy based on sliding spectrum decomposition (SSD) is proposed to determine the minimum recommended number of selected bands in different scenes. The superiority of the selected subset of bands is verified by random forest, support vector machine (SVM), and edge-preserving filtering-based SVM (EPF-SVM) classifiers. Experimental results on three well-known datasets demonstrate the robustness and superiority of the proposed HCS-MF algorithm compared with the state-of-the-art methods, such as marginalized graph self-representation (MGSR), neighborhood grouping normalized matched filter (NGNMF), and region-wise multiple graph fusion (RMGF).**

***Index Terms*— Band selection (BS), cuckoo search (CS) algorithm, hyperspectral image (HSI) classification, matched filter (MF), sliding spectrum decomposition (SSD).**

I. Introduction

With advances in spectral imaging technology, hyperspectral images (HSIs) captured by sensors featured abundant spectral information, which distinguishes subtle differences among land-cover materials easily [1]. Therefore, HSI is utilized in a variety of fields, such as medicine [2], agriculture [3], and architecture [4]. As the number of spectral channels increasing, the number of frequency bands increases accordingly, and hence brings a heavy burden to the research task of HSI classification [5], target detection [6], and change detection [5], due to the curse of the dimensionality problem [7]. To alleviate the high computation and storage requirement resulting from the explosive increase of hyperspectral data, it is critical to investigate dimensionality reduction (DR) techniques [8], [9].

Feature selection [10] and feature extraction [11] are the main research subjects of DR techniques for HSI data. The former is to select a suitable group of bands from the dataset by pre-setting criteria, also known as band selection (BS) [12]. The latter is to generate new features through a series of transformations from the original spectral bands [13]. Generally, HSI BS includes supervised [14], and unsupervised methods [15]. Given the prior knowledge, supervised BS methods [14] can achieve better DR performance while providing more accurate features. Unsupervised BS methods [15] rely on the assumption of the data to complete the DR, but the performance is usually not as good as the former. However, the supervised method often needs labels for the data to conduct dimension reduction, which limits its application. Due to the nature of the high complexity and large scale of the HSI, it is extremely time-consuming and labor-intensive to carry out the task of labeling, which is also prone to errors. Therefore, unsupervised BS methods have broader applicability, and this article will focus on unsupervised BS methods. Currently, swarm intelligent optimization algorithms are extensively adopted in the field of hyperspectral BS, e.g., particle swarm optimization (PSO) [16], genetic algorithm (GA) [17], and salp swarm algorithm (SSA) [18], by virtue of the advantages of fewer parameters and high search performance, etc. However, the above methods still suffer insufficient search ability in the late stage of the optimization search. Cuckoo search (CS) algorithm [19] is moreover one of the representative algorithms, which is also the research object of this article.

CS algorithm is characterized by satisfactory optimization performance both in global exploration and local exploitation phases. However, the lack of local exploitation capability in the Lévy flight and preference of random wandering strategies result in local extremes in the later phases of the algorithm [20]. The quantum mechanism has been demon-

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Meng Wu, Xianfeng Ou, Youli Lu, Wujing Li, Dan Yu, and Zhihao Liu are with the School of Information Science and Engineering, Hunan Institute of Science and Technology, Yueyang, Hunan 414006, China (e-mail: 1794909836@qq.com; ouxf@hnist.edu.cn; 151909445@qq.com; liwj@hnist.edu.cn; 56759907@qq.com; 2624573553@qq.com).

Chengtao Ji is with the School of Advanced Technology, Xi'an Jiaotong–Liverpool University, Suzhou 215123, China (e-mail: jichengtao@gmail.com).

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strated to be more robust to perturbation in the local search space, and thus can provide the CS algorithm with local search capability in the later iterations. Therefore, a heterogeneous CS (HCS) algorithm is proposed in this article. First, motivated by the band removal method of matched filter (MF) ranking [21], a BS optimization model is constructed based on MF to convert the BS problem into a combinatorial optimization problem. Then, a mapping method based on neighborhood band grouping (NBG) [22] is proposed to realize the mapping of the population to a sequence of bands, so as to realize the combination of the HCS algorithm and the BS optimization model (HCS-MF). Subsequently, A recommendation method for the number of target bands is proposed based on sliding spectrum decomposition (SSD) [23]. Finally, random forest, support vector machine (SVM), and edge-preserving filtering-based SVM (EPF-SVM) classifiers are applied for HSI classification to evaluate the efficacy of the subset of selected bands.

The most significant contributions of this article can be summarized as follows.

1) To enhance the search capability of the CS algorithm in the late iteration, the quantum mechanism is adopted to improve the Lévy flight and preference random wandering strategies, thereby a HCS algorithm is proposed.
2) A BS optimization model based on normalized MFs is proposed based on the relationship between MFs and the signal-to-noise ratio of each band.
3) A mapping method based on NBG is designed to transform the solution of HCS algorithm into candidate bands by taking into account that the similarity of adjacent bands is much higher than that of non-adjacent bands.
4) A recommendation method based on SSD for automatically deciding the number of target bands is designed through analyzing the quality and diversity of a subset of bands.

The rest of this article is organized as follows. Section II presents the related works about HSI BS. In Section III, the proposed HCS-MF algorithm is described in detail. Section IV gives extensive experimental results and analysis. In Section V, a conclusion is given.

II. Related Work

With respect to the generation of subset of bands, unsupervised BS methods are categorized into ranking-based [24], search-based [25], clustering-based [26], and point-by-point-based [27] methods.

A. Ranking-Based Method

The significance of each spectral band is quantified by a predefined criterion, and a top subset of bands are selected for further HSI research, such as information capacity, correlation, and dissimilarity. In [28], a method was proposed to quantify the utility of bands for classification based on mutual information (MI) to retain bands with high correlation MI values. In [29], a BS method for HSI classification was proposed based on self-MI and prominent bands. In [30], a method for quantifying the dissimilarity between bands was proposed to select a subset of bands with low similarity, drawing on the principle of MI. In [31], constrained multi-BS was extended with constrained BS-constrained energy minimization to achieve linearly constrained MV to constrain and select multiple bands simultaneously. In [32], a BS method based on band decorrelation was implemented using a local curve-fitting technique.

B. Searching-Based Method

The BS is formulated as an optimization problem, and the optimal or suboptimal band combinations can be obtained by applying the search strategy to this optimization problem. This article focuses on the research related to the combination of metaheuristic optimization algorithms with criterion functions. In [33], a feature selection method based on an improved ant-lion optimizer and a wavelet SVM was proposed. In [34], a BS method based on an improved sub-control decomposition and an artificial bee colony algorithm was proposed, which solved the DR problem in HSI classification by calculating the correlation coefficients between adjacent bands and implementing a subspace decomposition based on spectral curves. In [35], a BS method based on a whale optimization algorithm was proposed by a searching mechanism like the humpback whale hunting behavior and extracting intrinsic and edge space features through hybrid filtering. In [36], a multi-objective unsupervised BS model is designed based on band correlation and information, and hence a BS method based on an improved multi-objective artificial bee colony algorithm was proposed. In [37], a Lévy flight-based sine cosine BS method was proposed, and the band dimension of HSI was reduced using a substitution distribution. In [15], an unsupervised BS method based on an improved multi-objective CS algorithm was proposed based on the relationship between the amount of band information and correlation.

C. Clustering-Based Method

The method first performs a subspace partition (SP) and then selects bands from each group to form a subset of bands. In [38], the Ward's linkage method is proposed for BS based on MI and Kullback Leibler (KL) scatter, respectively, and it is also the earliest BS method using the clustering idea. In [39], K-means is used to realize clustering, and then bands are selected by traversing all clusters. However, K-means is a hard clustering, and a band may belong to different clusters at the same time, so the SP using K-means is not precise enough. In [40], a novel approach for hyperspectral BS via an adaptive SP strategy (ASPS) is proposed.

D. Point-by-Point-Based Method

The method obtains a subset of bands by continuously adding or removing bands to or from the result set, with each iteration reaching a local optimum and all affecting subsequent decisions accordingly [27]. In [41], a BS method based on linear prediction is proposed, which measures the degree of similarity between selected bands and candidate

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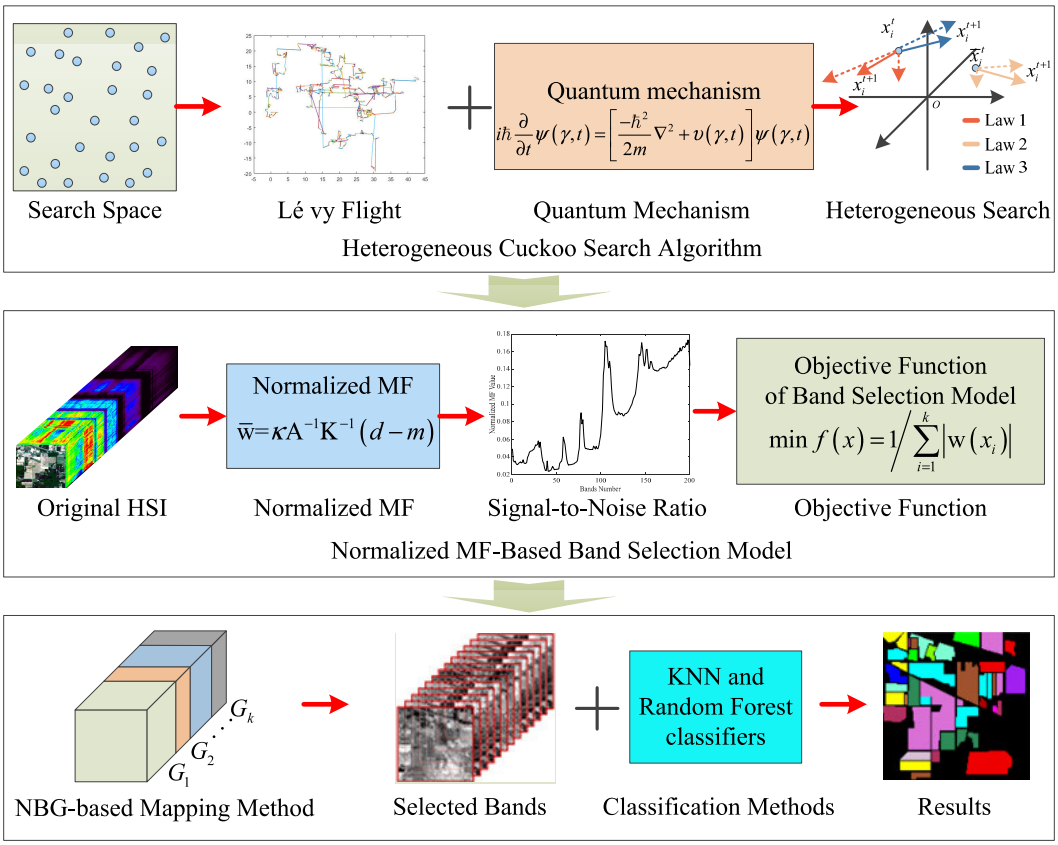


Fig. 1. Flowchart of the proposed HCS-MF method.

bands, and candidate bands with less similarity are susceptible to selection. In [42], a volume gradient BS method is proposed to describe the similarity between bands in terms of elliptical geometry and maximize the elliptical volume of candidate bands by removing concentrated bands one by one.

Although these methods obtained satisfactory performance, they ignored the correlation among bands and the sensitivity of the algorithms to bands with noise, which led to a possible high similarity between the selected bands. In addition, some of the algorithms are based on spatial transformations to reduce the dimensionality, which change the original HSI information. All the above can have some implications for HSI processing.

III. Proposed HCS-MF Algorithm

In this section, we will detail the proposed HCS-MF method and the corresponding flowchart is shown in Fig. 1. A novel MF-based BS model is designed which is based on the sensitivity of the MF to the band signal-to-noise ratio, and a HCS algorithm is proposed accordingly. In addition, an NBG-based mapping method is proposed to reduce the similarity and correlation among the selected bands. These three parts are described in detail in Sections III-A–III-C.

A. HCS Algorithm

1) Quantum Mechanism: The quantum computing here is applied to the CS algorithm to compensate for the effects of the lack of perturbation ability of Lévy flight and preference for random wandering in the late iterations. The change of the quantum state of the physical system against time is described in the following [43]:

$$i\hbar\frac{\partial}{\partial t}\psi(\gamma,t)=\left[\frac{-\hbar^2}{2m}\nabla^2+\upsilon(\gamma,t)\right]\psi(\gamma,t) \tag{1}$$

where i is imaginary unit, $\hbar$ is the reduced Planck constant, m represents particle's mass, ∇^2 is the Laplacian operator, and $\psi(\gamma,t)$ represents the wave function of the quantum system at position γ at time t.

Disregarding time t, the delta potential is the Schrödinger equation for the wave function $\psi(\gamma)$ of a particle in 1-D in a potential $\upsilon(\gamma)$, as shown in the following:

$$\upsilon(\gamma)=-\lambda\delta(\gamma) \tag{2}$$

where $\delta(\gamma)$ is the Dirac delta function, λ is the intensity of Dirac delta function. The solution of $\psi(\gamma)$ as shown in the following:

$$\psi(\gamma)=L^{-\frac{1}{2}}e^{-\frac{|\gamma|}{L}} \tag{3}$$

where $L=\hbar^2/m\gamma$.

The wave function $\psi(\gamma)$ represents the quantum state of each solution of CS algorithm. $|\psi(\gamma)|^2$ is the probability density function of the candidate solution, and as shown in the following:

$$|\psi(\gamma)|^2=L^{-1}e^{-\frac{2|\gamma|}{L}} \tag{4}$$

where $\eta=\exp(-2|\gamma|/L)$ is defined, then $|\gamma|=-L\ln(\eta)/2$.

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2) Heterogeneous Search: The CS algorithm consists of global exploration and local exploitation, the former of which is achieved by the Lévy flight for population updating as shown in the following:

$$x_i^{t+1} = x_i^t + \alpha \oplus \text{Lévy}(\lambda) \tag{5}$$

where x_i^t is the ith candidate solution of the tth iteration, $\alpha = 0.01$, $Lévy(\lambda) = \mu/|v|^{1/\beta}$, $\mu \sim N(0, \sigma_\mu^2)$, $v \sim N(0, 1)$, σ_μ as shown in the following:

$$\sigma_\mu = \left[\frac{\Gamma(1+\beta) \times \sin(\pi\beta/2)}{\Gamma\left[(1+\beta)/2\right] \times \beta \times 2^{(\beta-1)/2}}\right]^{1/\beta}. \tag{6}$$

The latter is achieved by preference random wandering to perform population updates as shown in the following:

$$x_i^{t+1} = x_i^t + r \cdot \left(x_j^t - x_k^t\right) \tag{7}$$

where x_i^t is the ith candidate solution of the tth iteration, $r \in \text{rand}(0, 1)$, x_j^t, and x_k^t represent the jth and kth solutions, respectively. However, the search capability is not sufficient to support the algorithm to obtain better performance in the late iteration.

Defining L as the distance between two candidate solutions and the average position $\bar{x}^t$ of all candidate solutions based on the quantum mechanism, $L = 2\delta|\bar{x}^t - x_i^t|$ and $s_i^t = |\gamma|$, then the new expression for the step size of the Lévy flight is in the following:

$$s_i^t = \delta|\bar{x}^t - x_i^t| \ln\left(\eta^{-1}\right) \tag{8}$$

where δ is the parameter that controls the difference between x_i^t and $\bar{x}^t$, $\bar{x}^t = {}^1\!/_n \sum_{i=1}^n x_i^t$.

With respect to the above descriptions, to enhance the perturbation capability of the CS algorithm in the search space, a heterogeneous search strategy is proposed on the basis of (5) and the quantum mechanism, as shown in the following:

$$x_i^{t+1} = \begin{cases} x_i^t + \alpha \oplus \text{Levy}(\lambda), & \text{sr} \in \left(2/31\right] \\ \bar{x}^t + \zeta\left[\bar{x}^t - x_i^t\right], & \text{sr} \in \left(1/32/3\right] \\ x_i^t + \varepsilon\left[g^t - x_i^t\right], & \text{else} \end{cases} \tag{9}$$

where $\zeta = \delta \ln(1/\eta)$, $\varepsilon = \delta \exp(\eta)$, g^t denotes optimal individual of the tth generation, sr as shown in the following:

$$\text{sr} = \frac{1}{\sqrt{2\pi}\sigma} e^{\left(-\left(x_i^t - \bar{x}^t\right)^2/2\sigma^2\right)} \tag{10}$$

where σ^2 is the variance of the whole population. Differently from other mechanisms, the quantum mechanism is embedded into the local exploitation of CS algorithm for further improving the search capability, as shown in the following:

$$x_i^{t+1} = \begin{cases} x_i^t + r \cdot \left(x_j^t - x_k^t\right), & \text{sr} \in \left(2/3, 1\right] \\ \bar{x}^t + \zeta\left[\bar{x}^t - x_i^t\right], & \text{sr} \in \left(1/3, 2/3\right] \\ x_i^t + \varepsilon\left[g^t - x_i^t\right], & \text{else.} \end{cases} \tag{11}$$

In addition, a HCS algorithm is proposed to improve the exploration and exploitation of the algorithm, and the detailed procedure of the HCS algorithm is shown in Algorithm 1.

Algorithm 1 Heterogeneous Cuckoo Search Algorithm

Input:
$iter$: Number of iterations
N: Size of population
Pa: Discovery probability
H: Hyperspectral cube

Output:
X: A selected band subset

Initialize the population;
Calculate the MF value for each band of H;
Compile the population by NBG-based mapping method;
Evaluate the population by normalized MF-based band selection model;
Update the optimal result;
while $t < iter$ **do**
  for $i = 1$ to N **do**
    Update the x_i via Eq.(9);
  end for
  Compile the population by NBG-based mapping method;
  Evaluate the population by normalized MF-based band selection model;
  Update the optimal result;
  for $i = 1$ to N **do**
    if $r < Pa$ **then**
      Update the x_i via Eq.(10);
    end if
  end for
  Compile the population by NBG-based mapping method;
  Evaluate the population by normalized MF-based band selection model;
  Update the optimal result;
  $t = t + 1$;
end while
Output optimal band subset.

B. Normalized MF-Based BS Model

Inspired by the method of removal of bad bands based on MF weights [21], a BS model based on normalized MF detector is proposed to select a subset of bands with higher signal-to-noise ratio. The weights calculated by MF detector represent the signal-to-noise ratio and the image quality of the bands, with the larger weights the greater signal-to-noise ratio. Therefore, an unsupervised BS model is designed as shown in the following:

$$\min f(x) = \frac{1}{\sum_{k=1}^K \sum_{i=1}^{\text{WH}} \frac{1}{\text{WH}} |w_i^{x_k}|} \tag{12}$$

where K represents the number of selected bands, x_k represents the selected band, W and H denote the width and height of the image, respectively. $w_i^{x_k}$ represents the weight of the ith pixel of the x_kth band obtained by the MF detector, as shown in the following:

$$w_i^{x_k} = \kappa C_{x_k}^{-1}\left(d_i^{x_k} - m^{x_k}\right) \tag{13}$$

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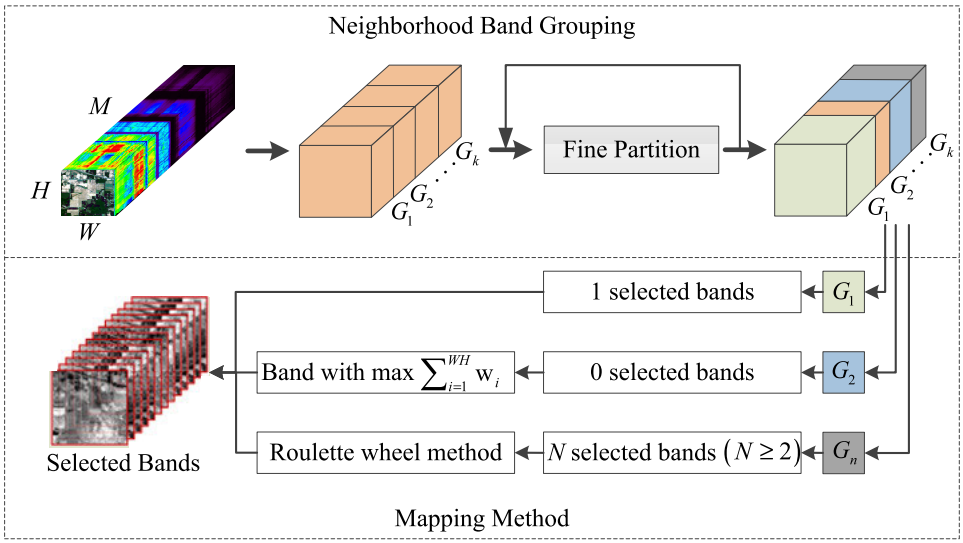


Fig. 2. NBG-based mapping method.

where C_{x_k} represents the covariance matrix of the x_kth band, $d_i^{x_k}$ represents the ith pixel point of the x_kth band, m^{x_k} represents the mean value of the x_kth band, and κ represents a normalization constant in the following:

$$\kappa = \frac{1}{\left(d_i^{x_k} - m^{x_k}\right)^T C_{x_k}^{-1}\left(d_i^{x_k} - m^{x_k}\right)}. \tag{14}$$

C. NBG-Based Mapping Method

In most cases, many scholars only consider the amount of information and noise contained in the bands and overlook the influence of correlation among bands on HSI classification. However, how to minimize the similarity among the selected bands is particularly significant. In general, the similarity between adjacent bands is usually higher than non-adjacent bands [22], thus resulting in lower similarity in the selected band subset by a novel mapping method design, as shown in Fig. 2.

In NBG method [22], the HSI cube is first divided into K groups as evenly as possible, and the point set G for partition is defined as shown in the following:

$$G_k = \begin{cases} 1, & k = 1 \\ ((M - \mathrm{mod}(M, K)) \times k)/K, & 2 \le k \le K - 1 \\ M, & k = K \end{cases} \tag{15}$$

where M is the number of bands, K is the number of groups and also is the number of selected bands, and $\mathrm{mod}(\cdot, \cdot)$ denotes mod operation. However, the lack of a theoretical basis for the initial division of the bands according to (15) resulted in an unsatisfactory result. Therefore, a fine-neighborhood grouping method is adopted to repartition G_k, where p_k is considered as the initial clustering center in each group, which is defined as shown in the following:

$$p_k = \left\lfloor k \times \frac{M}{K} - \frac{M}{2K} \right\rfloor, \quad k = 1, 2, \ldots, K. \tag{16}$$

To fully exploit the contextual information and reduce the execution time, only the bands that lie between a and b are taken into account, and (a, b) is defined by the following:

$$(a, b) = \begin{cases} [1, p_{k-1}), & k = 1 \\ (p_{k-1}, p_{k+1}), & 2 \le k \le K - 1 \\ (p_{k-1}, M], & k = K. \end{cases} \tag{17}$$

In the groups G_k partitioned by the NBG, the bands initially selected by the HCS algorithm are distributed in different groups, which can result in three different cases where there are 0, 1 or $\ge$ 2 selected bands in the groups G_k. For the selected bands in each group, the mapping method is performed with different strategies to get a subset of the selected bands respectively, as shown in Fig. 2. For only 0 candidate bands in the group, the band with the maximum matching filter weight is selected in the group; when there is 1 selected band in the group, the band is the final selected band; for $\ge$2 candidate bands in the group, the selection of the band is achieved by roulette selection strategy. To have similar probability that the initially candidate bands are mapped, a roulette strategy based on matching filter weights is designed in this article, which is defined as the following:

$$P_i = \sum_{j=1}^{i} \frac{\omega_j}{\sum_{l=1}^{L} \omega_l} \tag{18}$$

where L represents the number of bands in the group initially selected by the proposed HCS, $\omega = \sum_{i=1}^{\mathrm{WH}} w_i$. This operation reduce the similarity of the selected bands.

D. Recommended Bands From the Selected Bands

Presently, the BS methods consider more about the effect of different number of bands on classification accuracy, while how to determine the number of selected bands is less investigated. Therefore, we propose an SSD-based recommendation strategy for determining the number of bands.

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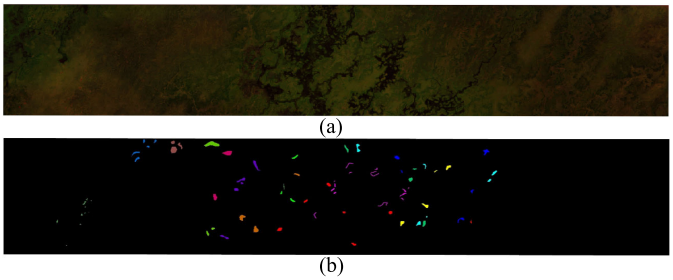


Fig. 3. Three-band color composing image and ground truth of Botswana. (a) Three-band color composite image. (b) Ground truth.

SSD mainly provides high-quality and diversified content sequences in content feeds [23], as shown in the following:

$$\max_{j \in Z \setminus I_{1:t}} r_j + \gamma \left\| T_{I_{1:t}}(v_j) \right\| \tag{19}$$

where γ is a hyperparameter balancing diversity and quality, and r and $T_{I_{1:t}}$ denote sequence quality and diversity, respectively.

To obtain the minimum number of selected bands, SSD measures the quality and diversity of the selected bands and the inflection point is the recommended number of selected bands. In this article, a smooth curve is obtained by fitting an exponential function to the discrete probability to obtain the inflection point of the curve to discriminate the recommended number of selected bands, as shown in the following:

$$r = \arg\max \sum_{i=1}^{K} \omega_i + \gamma \|T_i\| \tag{20}$$

where ω_i denotes the quality of the ith band, obtained from (21). T is obtained from (22)

$$\omega_i = \sum_{j=1}^{\mathrm{WH}} \left| w_j^i \right| \tag{21}$$

$$T_i = v_i - \sum_{j \neq i}^{K} \frac{\langle v_i, v_j \rangle}{\langle v_j, v_j \rangle} v_j \tag{22}$$

where v denotes the singular value of the ith band. Following the mentioned method, an optimal subset of recommended bands is finally obtained from the original data.

IV. Experimental Results and Analysis

A. Dataset Description

In the experiments, the public datasets, such as Botswana, Kennedy Space Center (KSC), and Salinas, are applied to validate the effectiveness and applicability of the proposed HCS-MF algorithm.

1) *Botswana:* The dataset was acquired by NASA EO-1 satellite in 2001 in the Okavango Delta, Botswana, and its size is 1476 × 256, which consists of 14 types of ground classes and 242 bands, as shown in Fig. 3. The actual number of bands used for training after removing the absorption zone and high noise bands is 145, containing seasonal swamps, occasional swamps, and dry woodlands.

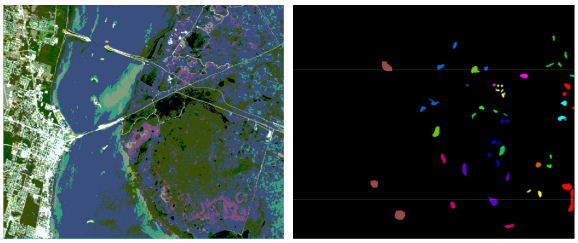


Fig. 4. Three-band color composing image and ground truth of KSC. (a) Three-band color composite image. (b) Ground truth.

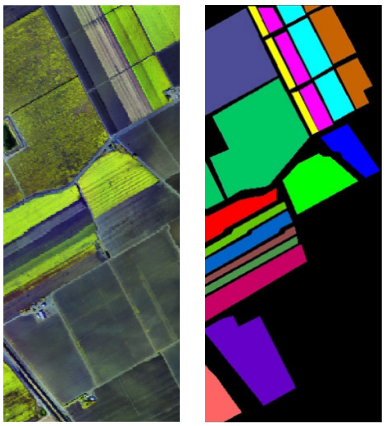


Fig. 5. Three-band color composing image and ground truth of Salinas. (a) Three-band color composite image. (b) Ground truth.

2) *Kennedy Space Center:* This dataset was acquired by the National Aeronautics and Space Administration using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 1996 in the KSC, Florida, and its size is 512 × 614, which consists 13 classes of land cover objects and 224 bands, as shown in Fig. 4. The wavelengths of bands range from 400 to 2500 nm. In addition, the pretreatment process removes 48 bands due to water spectral absorption, and 176 spectral bands are used in the experiment.
3) *Salinas:* This dataset was captured by the AVIRIS sensor in 1998 in the Salinas Valley, and it contains 16 classes of land cover objects and 224 bands. Each band image consists of 512 × 217 pixels. The wavelengths of bands range from 400 to 2500 nm. In addition, the pretreatment process removes 20 bands due to water spectral absorption, and 204 spectral bands are used in the experiment, as shown in Fig. 5.

B. Experimental Setup

To verify the quality of the selected bands, three classifiers, random forest, SVM, and EPF-SVM [44], are applied to classify the samples. Since all classifiers are supervised, we randomly selected 10% of each dataset as the training set. The same parameters are adopted for both classifiers on different datasets. There are many performance indicators in HSI classification to evaluate the classification results. In this article, the overall accuracy (OA) which is the percentage of correctly classified samples among all samples, the average

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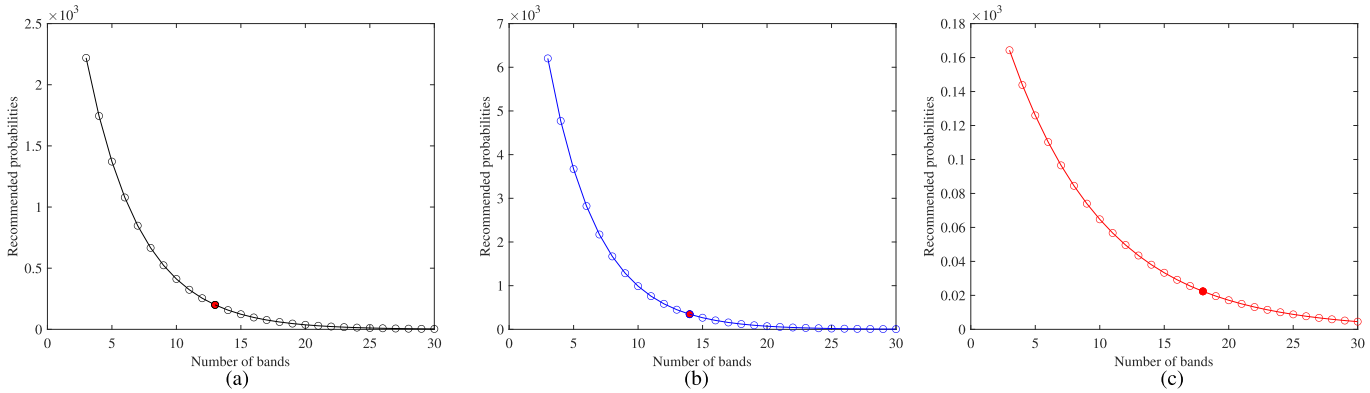


Fig. 6. Recommended probabilities of different selected band subsets. (a) Botswana. (b) KSC. (c) Salinas.

accuracy (AA) which is the average of the classification accuracy of each category, and Kappa coefficient which gives the percentage of correctly classified pixels, are employed. To verify the superiority of the proposed method, the following unsupervised BS methods are considered for comparison.

1) *ASPS [40]:* This method initially divides the HSI cube into multiple subcubes with the largest ratio of inter-class distance to intra-class distance. Then the band with the least noise was selected in each subcube.
2) *Maximum Variance Principal Component Analysis (MVPCA) [30]:* This method uses a type of information-theoretic criterion called divergence for dimension reduction, and BS was accomplished by combining band prioritization based on feature analysis with divergence-based band reduction.
3) *Robust Dual Graph Self-Representation (RDGSR) [45]:* This method uses the superpixel segmentation technique to generate homogeneous regions for each HSI to extract spatial information so that the superpixel-based similarity graph and the band-based similarity graph can be constructed.
4) *Neighborhood Grouping Normalized MF (NGNMF) [46]:* This method uses a band grouping strategy to reduce the redundancy among the original bands and an improved matching filter-based ranking strategy to reduce the effect of noisy bands on the classification results.
5) *Fast Neighborhood Grouping Method for Hyperspectral BS (FNGBS) [22]:* This method employs a coarse-fine strategy to divide the spatial HSI cube into several groups to mine contextual information in a large area. Then the most relevant and informative bands are obtained based on two factors: local density and information entropy.
6) *Marginalized Graph Self-Representation (MGSR) [47]:* This method uses superpixel segmentation of HSIs and recorded the relationship between neighboring pixels to obtain infinitely corrupted samples from the original pixels for model training. An alternative optimization algorithm was designed simultaneously to marginalize the corruption and search for the optimal solution.
7) *Region-Wise Multiple Graph Fusion (RMGF) [48]:* The method employs a superpixel segmentation method to partition the first principal component of the HSI into homogeneous regions and constructs a similarity graph for each superpixel to reflect the similarity between band pairs. Then, the spatial and spectral information of each superpixel is embedded in the graph diffusion process. Finally, the band containing the minimum noise in each subcube is selected to form a band subset.
8) *CS Algorithm With Match Filter (CS-MF):* This method uses the classic CS algorithm to optimize an objective function consistent with the HCS-MF method to select a subset of HSI bands.

To fairly verify the performance of the HCS-MF algorithm, the above methods are run independently for 30 times, where the parameters of the CS-MF and the HCS-MF algorithms are consistent across the experiment, such as the population size $N = 30$, the discovery probability $Pa = 0.25$. The method automatically terminates if the fitness of the algorithm is unchanged in five consecutive iterations. The number of bands selected is all 30. Bold fonts in all tables represent the best classification performance of all methods.

C. Recommended Bands Comparison

For determining the appropriate and reasonable number of selected bands, the recommendation probabilities curve shown in Fig. 6, which is obtained by the SSD-based band recommendation strategy. The number of recommended bands for the three datasets are 13, 14, and 18 respectively, and the selection results for different BS methods for the three datasets are shown in Table I. The MVPCA, RDGSR, and MGSR methods select a band subset consisting of a large number of many neighboring bands on the three datasets, whereas the high similarity between neighboring bands of the HSI indicates that the performance of the three BS methods is weak, which leads to unsatisfactory classification results. The other BS methods select bands that provide coverage of a larger range of bands, and the classification results of these methods are more satisfactory, as shown in Table II.

As summarized in Table II, the classification results for the subset of bands characterized by a wider distribution are more satisfactory. For the recommended number of selected bands

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TABLE I
RECOMMENDED SUBSET OF SELECTED BANDS OF THREE DATASETS

Algorithm	Band subset		
	Botswana (13 bands)	KSC (14 bands)	Salinas (18 bands)
ASPS	1,22,26,47,51,71,80,95,104,111, 131,133,145	1,25,34,52,53,77,91,93,97,111, 114,116,117,176	1,22,25,39,57,60,61,82,106,117,130, 142,150,156,178,182,201,204
MVPCA	29,30,31,33,34,35,36,39,40,41, 42,43,44	1,132,133,160,162,164,168,170,171,172, 173,174,175,176	41,42,43,44,45,46,47,48,49,50,51, 52,53,54,55,56,57,58
RDGSR	5,6,7,13,20,26,32,36,40, 41,43,46,54	47,69,70,74,75,76,83,84,115,139, 142,143,157,169	29,57,67,71,72,73,77,85,86,87,89, 102,116,125,129,132,135,143
NGNMF	6,14,23,34,36,48,58,78,86, 102,106,128,132	7,20,27,42,51,59,66,78,92,106, 126,128,155,159	6,17,29,30,52,55,72,74,92,116,120, 121,135,152,160,165,178,187
FNGBS	5,12,20,31,40,55,64,74,90, 97,115,123,137	1,16,23,42,49,62,67,80,92,117, 122,129,152,163	5,15,26,31,51,55,67,76,90,114,119, 125,133,150,165,168,183,193
MGSR	1,2,3,4,30,31,33,34,35, 36,40,43,44	1,132,133,156,160,167,168,169,170,171, 172,174,175,176	3,4,61,62,63,64,80,81,82,83,84, 85,96,98,102,103,104,105
RMGF	5,12,25,41,49,64,73,83,90, 99,119,133,139	8,18,32,38,50,65,67,70,87,104, 113,116,145,161	8,12,29,51,57,62,76,78,85,102,108, 116,124,135,157,179,186,200
CS-MF	4,17,18,35,37,57,71,81,82, 105,116,117,144	6,19,31,44,57,61,76,87,97,118, 127,128,157,176	1,17,28,40,51,60,64,83,106,107,119, 129,145,150,152,176,186,203
HCS-MF	5,17,28,35,44,57,72,81,82, 105,116,117,145	6,19,32,44,57,61,77,87,98,119, 127,132,157,176	1,17,27,40,51,63,73,83,106,107,117, 130,145,148,164,175,186,204

TABLE II
CLASSIFICATION ACCURACY OF PROPOSED HCS-MF AND OTHER STATE-OF-THE-ART ALGORITHMS

Dataset	Algorithm	Random Forest			SVM			EPF-SVM		
		OA(%)	AA(%)	Kappa(×100)	OA(%)	AA(%)	Kappa(×100)	OA(%)	AA (%)	Kappa(×100)
Botswana (13 bands)	ASPS	81.06+1.15	82.17+1.11	79.48+1.25	92.20+0.53	92.89+0.54	91.56+0.58	98.96+0.51	99.01+0.46	98.88+0.55
	MVPCA	67.91+1.90	68.45+1.96	65.21+2.06	92.57+0.55	93.15+0.51	91.95+0.59	97.61+0.40	97.91+0.31	97.41+0.44
	RDGSR	81.25+0.83	81.96+0.85	79.68+0.89	92.07+0.48	92.82+0.45	91.41+0.52	98.79+0.28	98.92+0.24	98.69+0.31
	NGNMF	81.26+1.39	82.13+1.47	79.69+1.50	93.04+0.50	93.73+0.47	92.47+0.54	98.23+0.45	98.44+0.36	98.08+0.48
	FNGBS	84.04+0.60	85.07+0.74	82.72+0.65	93.14+0.49	93.74+0.45	92.57+0.53	98.22+0.45	98.37+0.37	98.07+0.48
	MGSR	75.92+0.82	76.08+1.18	73.91+0.90	92.38+0.72	93.01+0.71	91.74+0.78	98.32+0.36	98.43+0.30	98.18+0.39
	RMGF	82.92+0.86	84.05+0.84	81.50+0.93	92.09+0.75	92.86+0.66	91.43+0.81	98.11+0.31	98.27+0.28	97.95+0.34
	CS-MF	84.26+0.81	85.09+0.91	82.95+0.87	92.64+0.83	93.29+0.72	92.03+0.90	98.69+0.57	98.82+0.52	98.58+0.62
	HCS-MF	**85.36+0.48**	**86.25+0.45**	**84.14+0.52**	**93.50+0.51**	**94.08+0.42**	**92.96+0.56**	**98.97+0.29**	**99.01+0.30**	**98.88+0.31**
KSC (14 bands)	ASPS	83.82+0.43	77.30+0.85	81.97+0.48	92.75+0.25	89.92+0.45	91.93+0.28	99.05+0.22	98.76+0.28	98.94+0.25
	MVPCA	64.23+0.56	52.57+0.89	59.94+0.61	67.77+0.52	58.83+1.07	63.82+0.59	81.01+2.77	82.17+6.26	78.61+3.14
	RDGSR	77.02+0.40	67.86+0.77	74.33+0.45	87.89+0.42	83.92+0.73	86.48+0.47	95.97+1.18	97.32+0.69	95.50+1.32
	NGNMF	84.65+0.58	76.71+0.90	82.89+0.64	93.64+0.47	90.62+0.68	92.91+0.53	99.07+0.26	98.82+0.32	98.97+0.29
	FNGBS	84.40+0.65	77.49+0.84	82.82+0.73	94.10+0.24	91.36+0.36	93.43+0.26	99.17+0.21	98.92+0.25	99.08+0.23
	MGSR	66.66+0.65	55.65+0.71	62.70+0.71	83.63+0.78	78.42+1.05	81.71+0.88	93.95+0.96	96.46+0.59	93.23+1.08
	RMGF	84.53+0.89	77.32+0.97	82.75+0.99	93.69+0.27	90.47+0.61	92.97+0.30	99.26+0.25	99.01+0.30	99.18+0.28
	CS-MF	84.32+1.14	77.14+1.67	82.52+1.27	93.63+0.61	90.60+1.04	92.90+0.67	99.52+0.33	99.39+0.42	99.47+0.36
	HCS-MF	**84.69+0.35**	**77.67+1.00**	**82.93+0.40**	**94.15+0.26**	**91.49+0.37**	**93.48+0.29**	**99.56+0.17**	**99.43+0.23**	**99.51+0.19**
Salinas (18 bands)	ASPS	90.35+0.12	94.44+0.12	89.25+0.13	90.85+0.24	95.93+0.23	89.6+0.26	96.34+0.72	98.47+0.22	95.83+0.82
	MVPCA	82.91+0.16	85.42+0.37	80.94+0.18	89.47+0.24	93.42+0.28	88.04+0.27	95.77+0.39	98.13+0.24	95.19+0.44
	RDGSR	89.51+0.20	93.43+0.26	88.32+0.22	91.53+0.23	96.07+0.27	90.37+0.26	96.46+0.74	98.52+0.26	95.97+0.84
	NGNMF	90.81+0.10	94.79+0.07	89.76+0.12	91.57+0.18	96.16+0.26	90.41+0.20	96.37+0.78	98.50+0.23	95.87+0.89
	FNGBS	90.59+0.19	94.70+0.24	89.52+0.21	91.58+0.13	96.05+0.22	90.43+0.15	96.37+0.68	98.47+0.20	95.87+0.78
	MGSR	87.74+0.14	91.03+0.27	86.35+0.16	89.30+0.17	93.84+0.23	87.84+0.19	96.21+0.65	98.46+0.20	95.69+0.74
	RMGF	90.85+0.19	94.82+0.22	89.81+0.21	90.76+0.49	95.72+0.32	89.50+0.56	96.56+0.59	98.59+0.13	96.08+0.67
	CS-MF	90.63+0.25	94.57+0.16	89.57+0.27	91.01+0.51	95.84+0.31	89.77+0.57	96.28+0.84	98.53+0.31	95.76+0.95
	HCS-MF	**90.99+0.15**	**94.88+0.12**	**89.97+0.16**	**91.66+0.30**	**96.20+0.17**	**90.52+0.34**	**97.15+0.39**	**98.84+0.18**	**96.75+0.45**

for the three datasets, the proposed HCS-MF algorithm outperforms the other comparison methods for the classification results obtained with different classifiers.

D. Botswana

The numerical evaluation on classification results of the proposed HCS-MF method and the other methods on different classifiers for the selected band subset are shown in Tables III–V respectively, which include the OA, AA, and Kappa coefficients, and the classification accuracy for each class. Fig. 7 shows the classification results of the HCS-MF method and the other methods on different classifiers when different numbers of bands are selected.

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TABLE III

CLASSIFICATION ACCURACY OF PROPOSED HCS-MF AND OTHER STATE-OF-THE-ART ALGORITHMS ON BOTSWANA BY RANDOM FOREST

Class	ASPS	MVPCA	RDGSR	NGNMF	FNGBS	MGSR	RMGF	CS-MF	HCS-MF
1	99.04±1.33	99.22±0.71	99.07±1.01	**99.70±0.29**	98.81±2.21	98.85±1.22	98.33±2.43	97.70±2.53	99.04±1.65
2	**88.71±7.78**	67.72±9.79	73.17±8.04	83.56±10.26	86.24±5.59	66.53±11.48	79.90±4.98	83.07±9.22	87.33±8.79
3	**91.12±2.73**	82.99±4.76	85.38±4.28	88.53±6.27	89.92±4.08	87.49±3.23	88.49±4.18	88.65±6.11	89.84±5.73
4	86.28±4.35	84.60±4.39	82.37±2.51	89.58±5.37	85.07±5.17	83.63±3.94	84.19±5.60	86.33±4.33	**89.95±1.84**
5	77.81±2.26	73.12±6.31	75.24±1.79	74.68±6.14	76.84±5.12	76.28±4.27	76.51±6.63	78.85±3.88	**80.04±2.87**
6	59.96±6.13	57.77±7.03	55.80±2.51	53.38±2.79	62.23±6.82	53.49±4.61	63.31±7.45	56.28±3.67	**63.98±5.43**
7	95.14±2.68	91.54±4.32	93.36±3.84	94.17±4.62	94.52±3.73	89.92±4.29	95.29±2.57	92.43±3.75	**96.41±1.45**
8	89.51±3.44	78.97±8.47	82.32±4.04	81.87±4.36	**90.39±3.02**	84.24±3.77	87.54±7.38	86.45±5.70	89.66±4.10
9	77.74±5.05	73.18±5.37	72.68±4.41	72.07±7.27	76.24±6.77	73.63±3.50	76.40±3.92	74.49±4.33	**79.55±4.81**
10	80.56±4.96	78.43±4.41	80.12±4.43	71.77±5.46	79.80±6.29	71.98±5.23	76.25±5.13	77.82±5.63	**81.57±3.19**
11	**91.05±2.34**	85.21±3.84	83.08±4.17	86.13±5.20	89.90±3.50	87.77±3.78	90.33±3.88	90.98±2.92	90.95±3.22
12	89.67±5.97	**92.76±3.85**	89.01±4.27	92.04±3.86	91.33±4.50	89.01±2.04	90.17±4.88	89.83±2.80	91.71±2.94
13	79.14±8.20	78.25±4.73	81.08±4.55	78.17±4.22	82.35±2.26	79.48±5.69	80.75±5.24	79.93±5.82	**84.55±5.40**
14	97.26±2.23	99.05±1.68	98.95±0.70	97.89±2.16	96.32±2.82	**99.47±0.55**	95.26±3.95	97.05±2.09	97.79±2.78
OA(%)	84.92±0.97	81.03±1.06	81.49±0.71	81.87±0.95	84.79±0.19	81.05±0.72	83.88±0.71	83.36±0.81	**86.47±0.46**
AA(%)	85.93±0.84	81.63±1.00	82.26±0.82	83.11±0.94	85.71±0.35	81.56±0.96	84.48±0.71	84.28±1.02	**87.31±0.53**
Kappa(×100)	83.67±1.05	79.45±1.15	79.95±0.76	80.36±1.02	83.52±0.20	79.46±0.78	82.54±0.77	81.98±0.88	**85.35±0.50**

TABLE IV

CLASSIFICATION ACCURACY OF PROPOSED HCS-MF AND OTHER STATE-OF-THE-ART ALGORITHMS ON BOTSWANA BY SVM

Class	ASPS	MVPCA	RDGSR	NGNMF	FNGBS	MGSR	RMGF	CS-MF	HCS-MF
1	99.85±0.31	99.85±0.31	99.78±0.36	99.63±0.52	99.85±0.31	**100.0±0.00**	99.93±0.23	99.93±0.23	99.93±0.23
2	96.57±1.45	95.88±2.48	97.30±2.06	95.52±2.42	95.48±1.24	95.95±1.60	95.55±2.68	95.37±2.72	**97.92±2.36**
3	98.05±1.45	96.69±1.36	93.83±2.73	**99.68±0.41**	98.82±0.76	94.59±1.76	98.21±1.27	97.70±2.37	98.67±0.97
4	94.29±2.55	91.52±2.71	91.82±2.93	**96.49±2.19**	94.96±1.14	92.75±2.17	94.57±1.90	93.62±2.45	96.07±1.61
5	89.80±2.36	82.75±2.02	87.09±2.27	89.05±1.78	89.27±2.06	84.95±3.03	87.45±3.16	88.00±1.56	**91.39±2.06**
6	84.28±1.46	79.01±1.50	80.28±2.75	83.22±2.61	81.53±3.05	78.16±3.10	80.31±2.82	80.59±1.79	**85.33±3.71**
7	99.92±0.25	99.69±0.54	99.00±0.82	99.92±0.24	99.69±0.74	99.46±0.72	**99.92±0.24**	99.54±0.82	99.69±0.54
8	**98.55±1.12**	97.57±1.43	95.83±1.97	97.52±1.42	96.72±1.35	96.67±1.31	97.03±1.43	97.05±1.51	98.25±0.76
9	90.87±2.60	**94.92±1.68**	90.30±3.02	86.92±2.00	90.71±2.37	92.13±2.85	91.92±2.46	90.56±3.07	93.17±2.01
10	90.10±2.50	88.50±1.36	90.54±2.50	86.42±3.22	91.00±2.18	88.27±3.09	90.87±2.80	89.53±3.42	**93.30±1.81**
11	95.57±2.02	91.87±2.26	92.87±2.41	92.25±1.77	95.53±1.47	91.30±1.21	95.20±2.31	94.67±1.89	**95.59±1.59**
12	93.38±2.35	**96.43±1.46**	94.95±1.94	94.50±2.41	94.52±2.55	93.06±1.62	93.53±2.34	92.79±3.17	94.46±1.63
13	94.14±1.73	91.61±3.04	93.94±1.45	94.38±1.89	94.85±1.87	93.70±2.70	94.35±1.69	94.57±2.14	**95.13±1.65**
14	99.80±0.65	99.18±1.05	98.76±1.07	97.95±0.94	99.18±1.06	**100.0±0.00**	99.80±0.65	99.59±0.86	99.80±0.65
OA(%)	94.16±0.43	92.60±0.50	92.65±0.48	93.24±0.52	93.98±0.07	92.20±0.53	93.67±0.49	93.30±0.25	**95.16±0.24**
AA(%)	94.66±0.46	93.25±0.45	93.31±0.39	93.82±0.50	94.44±0.15	92.93±0.47	94.19±0.40	93.82±0.33	**95.62±0.25**
Kappa(×100)	93.67±0.46	91.98±0.55	92.04±0.52	92.68±0.56	93.48±0.08	91.56±0.57	93.14±0.53	92.75±0.27	**94.76±0.26**

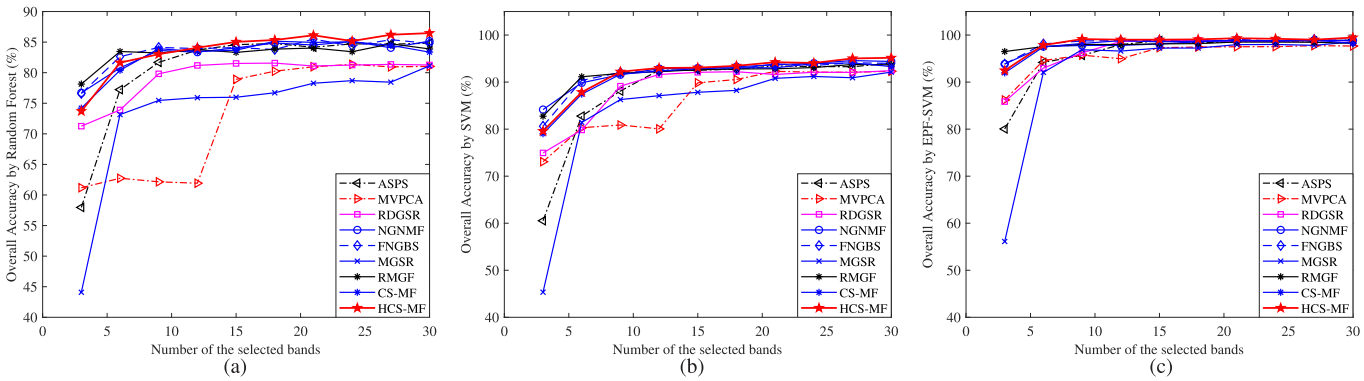


Fig. 7. Results by selecting different number of bands on Botswana. (a) Random forest. (b) SVM. (c) EPF-SVM.

As can be seen from Table III, with random forest as the classifier, the proposed HCS-MF algorithm has the highest classification accuracy in seven out of 14 classes and the proposed method has optimal OA, AA, and Kappa coefficients when compared to the other methods. In addition to the proposed method, the highest classification accuracy is achieved by ASPS with an OA of 84.92% and the worst is MVPCA with an OA of 81.03%. With SVM as the classifier in Table IV, the

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TABLE V
CLASSIFICATION ACCURACY OF PROPOSED HCS-MF AND OTHER STATE-OF-THE-ART ALGORITHMS ON BOTSWANA BY EPF-SVM

Class	ASPS	MVPCA	RDGSR	NGNMF	FNGBS	MGSR	RMGF	CS-MF	HCS-MF
1	99.56±0.38	99.78±0.36	99.56±0.38	99.49±0.36	99.78±0.36	99.49±0.36	99.56±0.38	**99.93±0.23**	99.49±0.36
2	**100.0±0.00**	98.67±1.56	**100.0±0.00**	99.81±0.61	99.25±1.59	98.46±1.77	99.05±1.61	98.87±1.82	99.43±1.28
3	**100.0±0.00**	99.77±0.74	99.14±0.58	**100.0±0.00**	**100.0±0.00**	99.06±0.72	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
4	93.04±3.25	94.59±1.22	**96.57±2.38**	95.68±1.52	89.80±1.51	94.91±3.05	91.68±2.80	89.32±2.59	95.96±2.11
5	97.74±3.48	87.72±2.61	94.98±3.13	96.23±1.96	98.01±1.22	91.43±2.16	95.38±3.12	95.72±4.45	**98.94±0.88**
6	**100.0±0.00**	99.83±0.35	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	99.68±0.55	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
7	**100.0±0.00**	**100.0±0.00**	99.70±0.53	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
8	99.90±0.31	99.51±0.51	99.03±0.79	98.84±0.75	**100.0±0.00**	99.04±1.01	100.0±0.00	99.44±1.48	99.90±0.31
9	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	96.99±1.75	99.94±0.20	99.17±2.40	99.94±0.20	99.75±0.79	**100.0±0.00**
10	96.17±2.10	96.84±2.54	98.03±1.05	94.48±3.58	99.92±0.25	95.71±2.60	94.54±3.29	96.96±2.94	**99.53±0.99**
11	99.93±0.21	99.73±0.65	99.74±0.62	99.54±0.54	99.87±0.27	99.13±0.55	**100.0±0.00**	99.93±0.21	99.81±0.31
12	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
13	98.68±1.07	98.18±1.18	97.53±0.83	98.69±1.2	97.97±1.15	98.19±1.57	97.13±1.74	96.52±2.18	**99.41±0.68**
14	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
OA(%)	98.84±0.45	97.97±0.31	98.76±0.42	98.35±0.51	98.85±0.10	98.00±0.55	98.24±0.26	98.19±0.19	**99.46±0.14**
AA(%)	98.93±0.38	98.19±0.25	98.88±0.38	98.55±0.42	98.89±0.14	98.16±0.49	98.38±0.23	98.32±0.21	**99.46±0.15**
Kappa(×100)	98.74±0.48	97.80±0.34	98.66±0.46	98.22±0.56	98.76±0.11	97.83±0.59	98.10±0.29	98.04±0.21	**99.41±0.15**

TABLE VI
CLASSIFICATION ACCURACY OF PROPOSED HCS-MF AND OTHER STATE-OF-THE-ART ALGORITHMS ON KCS BY RANDOM FOREST

Class	ASPS	MVPCA	RDGSR	NGNMF	FNGBS	MGSR	RMGF	CS-MF	HCS-MF
1	94.02±1.82	91.60±1.74	86.41±1.78	94.13±1.39	**94.66±1.96**	89.97±2.41	92.93±2.34	94.28±1.65	94.32±2.07
2	77.98±6.55	51.65±6.90	79.05±5.77	81.56±4.45	83.74±2.89	71.89±2.94	82.43±7.92	79.92±4.32	**85.43±3.84**
3	86.33±3.55	65.55±4.57	65.12±5.91	86.52±3.07	88.09±2.95	75.78±6.61	83.98±3.77	**88.95±2.80**	88.32±3.19
4	62.90±7.21	35.91±5.84	44.37±3.51	64.29±5.05	64.64±4.77	50.52±4.25	62.78±5.93	55.28±7.98	**66.31±8.53**
5	55.16±4.97	27.45±4.81	47.27±5.18	54.10±4.85	52.55±5.66	44.22±5.83	51.80±5.16	50.87±7.17	**56.46±4.98**
6	50.39±5.71	32.27±4.53	40.70±5.53	47.60±5.82	46.81±4.35	41.88±3.21	41.53±6.83	45.94±8.54	**47.95±6.13**
7	72.67±8.03	42.76±4.17	67.71±10.58	66.95±8.91	**76.67±6.37**	51.62±5.68	75.81±10.32	73.33±10.13	71.81±7.24
8	75.38±5.87	40.16±3.73	64.78±4.99	75.03±4.25	75.22±3.47	59.23±5.85	72.18±5.67	74.15±6.10	**78.45±6.02**
9	90.19±4.75	84.31±4.74	84.08±3.43	90.08±4.34	91.48±2.62	86.85±2.96	89.58±4.27	86.98±5.06	**91.73±3.12**
10	85.47±4.84	23.59±2.39	84.55±3.35	**85.45±3.46**	85.30±3.78	74.53±3.84	83.34±4.82	82.05±4.09	84.41±3.65
11	93.87±2.04	94.92±1.80	95.18±1.79	96.40±2.23	95.06±2.22	96.21±1.86	94.39±2.59	96.13±1.98	**96.52±1.92**
12	84.83±3.26	58.23±2.60	81.31±4.06	85.39±2.37	**86.24±2.54**	79.01±3.81	83.76±3.00	83.18±3.66	84.23±3.91
13	99.55±0.41	96.94±1.75	99.40±0.44	99.58±0.42	99.46±0.43	**99.64±0.29**	99.23±0.75	99.48±0.44	99.58±0.55
OA(%)	85.66±0.89	67.93±0.77	79.75±0.56	85.55±0.49	86.03±0.16	79.34±0.78	84.23±0.74	84.24±0.27	**86.39±0.38**
AA(%)	79.94±1.21	57.33±0.89	72.30±1.13	79.01±1.01	79.99±0.53	70.87±0.98	77.98±1.10	77.73±0.76	**80.42±0.61**
Kappa(×100)	84.02±1.00	64.09±0.87	77.41±0.63	83.89±0.56	84.43±0.18	76.96±0.86	82.42±0.83	82.43±0.31	**84.83±0.41**

TABLE VII
CLASSIFICATION ACCURACY OF PROPOSED HCS-MF AND OTHER STATE-OF-THE-ART ALGORITHMS ON KSC BY SVM

Class	ASPS	MVPCA	RDGSR	NGNMF	FNGBS	MGSR	RMGF	CS-MF	HCS-MF
1	94.39±1.08	70.83±6.45	87.75±2.48	94.45±0.81	94.18±1.07	74.43±0.94	94.50±1.02	93.82±1.03	**94.97±0.80**
2	94.77±1.89	53.04±3.33	92.95±2.98	95.52±2.18	93.52±2.34	84.41±3.15	94.68±1.86	91.84±3.00	**95.56±1.77**
3	**93.40±1.97**	38.49±8.79	88.01±3.54	93.01±1.75	91.51±2.36	60.92±3.31	92.11±2.32	91.56±2.50	93.24±1.76
4	76.97±4.38	35.12±4.48	62.66±3.75	74.84±3.18	77.90±2.81	50.12±4.39	77.20±3.63	72.16±2.10	**78.97±3.56**
5	85.06±3.57	48.11±5.69	74.80±4.77	84.16±6.97	**85.99±2.56**	69.53±6.79	83.61±5.05	84.27±2.67	84.45±4.30
6	**87.49±2.52**	30.27±10.0	67.98±3.73	83.14±4.84	86.80±3.59	55.75±3.85	83.00±4.43	80.10±5.27	86.19±3.37
7	86.53±4.49	61.42±7.02	85.10±3.05	87.46±4.05	86.78±4.24	77.24±5.37	85.73±4.79	84.92±5.26	**88.17±2.47**
8	91.86±1.97	57.40±4.64	88.83±1.41	93.51±1.60	94.06±2.42	73.94±2.59	93.81±1.90	91.79±2.40	**94.39±0.84**
9	95.54±0.48	72.69±3.05	94.17±1.58	95.94±0.99	95.86±1.38	88.36±1.25	95.66±1.05	94.74±1.33	**96.30±0.86**
10	99.95±0.16	47.10±4.35	99.56±0.98	99.90±0.32	99.85±0.24	97.69±0.93	**100.0±0.00**	99.65±0.48	99.90±0.21
11	98.68±0.82	98.52±1.38	98.75±2.19	99.28±0.56	98.81±0.68	**99.42±0.92**	99.24±1.10	99.19±0.74	**99.42±0.38**
12	98.53±0.78	72.94±2.16	97.32±0.76	98.25±0.95	**98.85±0.86**	86.33±2.65	98.65±0.71	98.09±0.95	98.62±0.72
13	99.98±0.07	83.62±2.65	99.98±0.07	**100.0±0.00**	**100.0±0.00**	99.70±0.38	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
OA(%)	94.80±0.45	68.20±1.96	91.10±0.55	94.69±0.41	94.93±0.09	83.30±0.67	94.75±0.42	93.77±0.23	**95.35±0.37**
AA(%)	92.55±0.63	59.20±2.46	87.53±0.79	92.27±0.99	92.62±0.37	78.30±1.07	92.17±0.89	90.93±0.4	**93.09±0.73**
Kappa(×100)	94.20±0.50	64.30±2.25	90.08±0.62	94.09±0.45	94.36±0.10	81.34±0.75	94.15±0.46	93.06±0.25	**94.82±0.41**

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TABLE VIII
CLASSIFICATION ACCURACY OF PROPOSED HCS-MF AND OTHER STATE-OF-THE-ART ALGORITHMS ON KSC BY EPF-SVM

Class	ASPS	MVPCA	RDGSR	NGNMF	FNGBS	MGSR	RMGF	CS-MF	HCS-MF
1	**100.0±0.00**	67.48±11.55	98.61±1.16	**100.0±0.00**	**100.0±0.00**	75.24±2.98	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
2	99.84±0.34	78.27±8.63	99.51±0.42	**99.92±0.26**	99.84±0.34	98.86±0.68	99.76±0.39	99.59±0.43	99.76±0.39
3	100.0±0.00	73.91±15.69	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	94.92±4.24	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
4	93.03±4.83	93.99±6.87	88.88±2.79	90.40±1.98	95.55±1.74	89.32±7.15	91.61±3.58	86.11±1.32	**97.69±1.16**
5	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	99.64±1.15	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
6	100.0±0.00	80.00±42.16	99.5±1.57	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	99.83±0.54	**100.0±0.00**	**100.0±0.00**
7	100.0±0.00	99.68±1.02	100.0±0.00	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
8	98.68±0.50	77.49±8.63	97.43±0.80	98.95±0.64	98.86±0.80	95.02±1.92	98.37±0.74	98.41±0.74	**99.18±0.60**
9	99.92±0.16	73.51±6.67	**100.0±0.00**	99.85±0.20	99.96±0.12	99.92±0.16	**100.0±0.00**	99.81±0.20	**100.0±0.00**
10	**100.0±0.00**	88.52±5.38	99.61±0.93	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	99.95±0.16	**100.0±0.00**
11	**100.0±0.00**	**100.0±0.00**	99.63±1.16	99.81±0.40	99.81±0.40	**100.0±0.00**	**100.0±0.00**	99.81±0.45	99.86±0.45
12	99.84±0.38	86.47±4.76	99.88±0.27	99.96±0.12	**100.0±0.00**	98.38±2.44	**100.0±0.00**	99.88±0.37	**100.0±0.00**
13	**100.0±0.00**	93.18±0.95	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	99.89±0.23	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
OA(%)	99.49±0.29	81.90±4.39	98.86±0.30	99.36±0.11	99.65±0.05	93.85±0.79	99.39±0.17	99.02±0.12	**99.79±0.05**
AA(%)	99.33±0.38	85.58±4.59	98.70±0.30	99.15±0.14	99.54±0.09	96.25±0.57	99.20±0.25	98.74±0.13	**99.73±0.09**
Kappa(×100)	99.43±0.32	79.62±4.97	98.73±0.33	99.29±0.12	99.61±0.06	93.11±0.89	99.33±0.19	98.91±0.14	**99.77±0.06**

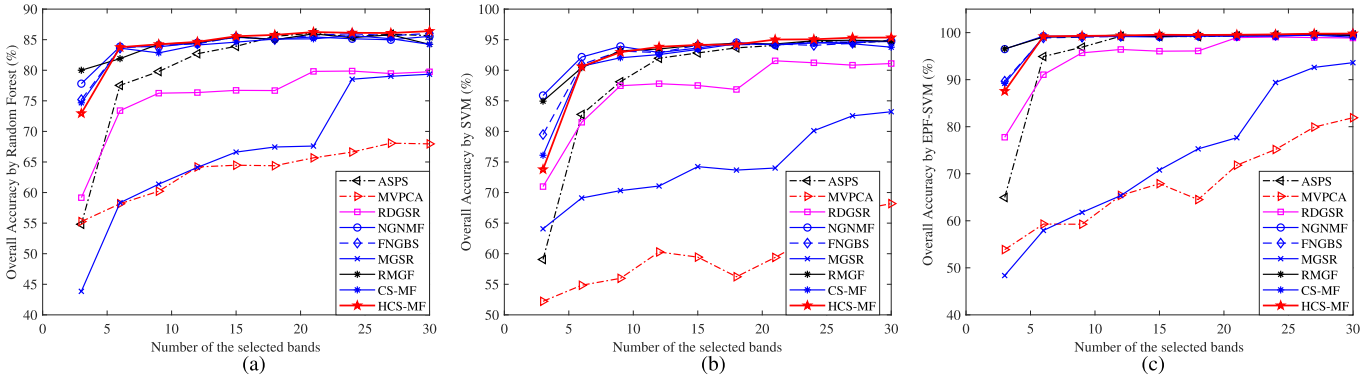


Fig. 8. Results by selecting different number of bands on KSC. (a) Random forest. (b) SVM. (c) EPF-SVM.

TABLE IX
CLASSIFICATION ACCURACY OF PROPOSED HCS-MF AND OTHER STATE-OF-THE-ART ALGORITHMS ON SALINAS BY RANDOM FOREST

Class	ASPS	MVPCA	RDGSR	NGNMF	FNGBS	MGSR	RMGF	CS-MF	HCS-MF
1	99.26±0.24	96.44±1.05	99.21±0.19	99.31±0.31	99.44±0.19	98.96±0.28	99.31±0.22	**99.56±0.12**	99.47±0.23
2	99.74±0.18	98.51±0.71	99.35±0.23	**99.80±0.05**	99.69±0.15	99.08±0.26	99.75±0.20	99.65±0.13	99.79±0.14
3	**98.64±0.64**	94.89±1.64	97.10±0.82	98.64±0.56	98.66±0.56	96.77±1.25	97.92±0.78	98.38±0.69	98.29±0.60
4	99.37±0.35	99.10±0.39	99.28±0.58	99.10±0.50	99.32±0.30	99.19±0.42	**99.40±0.27**	98.87±0.82	99.27±0.46
5	98.17±0.41	95.04±0.71	96.71±0.59	98.27±0.61	**98.28±0.42**	96.63±0.58	98.22±0.36	98.25±0.28	**98.28±0.38**
6	99.68±0.15	99.35±0.23	99.52±0.20	99.63±0.15	99.66±0.13	99.31±0.15	99.69±0.19	99.62±0.22	**99.75±0.07**
7	99.50±0.14	98.43±0.30	99.26±0.19	99.47±0.12	99.40±0.16	99.27±0.09	**99.50±0.18**	99.34±0.21	99.38±0.16
8	84.20±0.73	79.36±0.90	81.89±1.06	83.74±0.75	84.43±0.85	80.49±0.90	83.66±0.63	83.55±0.81	**84.62±0.60**
9	99.06±0.20	96.30±0.65	98.00±0.18	**99.33±0.15**	99.20±0.24	97.77±0.28	99.13±0.12	99.06±0.20	99.18±0.28
10	91.74±0.89	85.98±1.42	90.68±0.53	92.29±0.66	92.44±1.32	89.18±0.71	92.14±0.50	91.79±1.13	**92.65±0.59**
11	93.61±1.93	85.83±2.49	89.08±2.72	**96.03±0.73**	94.37±1.58	82.92±2.66	94.14±1.78	94.13±1.52	95.08±1.34
12	99.20±0.62	98.84±0.68	98.73±0.54	99.27±0.24	99.11±0.43	97.30±0.92	99.02±0.54	98.95±0.43	**99.33±0.42**
13	97.91±0.74	96.05±2.42	**98.31±0.97**	97.53±0.55	97.99±0.88	97.36±0.74	98.23±0.64	98.24±0.51	97.83±0.72
14	94.27±1.10	93.78±1.44	92.15±1.45	94.11±0.91	**94.08±1.29**	91.12±1.26	93.07±0.7	93.56±1.33	93.66±1.12
15	69.10±0.76	63.72±0.79	66.31±1.75	69.58±0.91	**69.51±1.35**	64.44±1.36	68.02±1.29	67.74±1.24	69.41±0.82
16	97.47±0.53	92.98±1.37	97.42±0.44	97.75±0.75	97.65±0.50	96.92±0.67	97.95±0.38	98.07±0.45	**98.20±0.32**
OA(%)	91.31±0.16	87.97±0.19	89.94±0.15	91.39±0.16	91.49±0.04	88.99±0.14	91.06±0.08	90.98±0.04	**91.55±0.09**
AA(%)	95.06±0.19	92.16±0.22	93.94±0.18	95.24±0.16	95.20±0.09	92.92±0.27	94.95±0.17	94.92±0.16	**95.26±0.07**
Kappa(×100)	90.32±0.17	86.60±0.21	88.79±0.17	90.41±0.18	90.51±0.04	87.73±0.16	90.04±0.09	89.95±0.04	**90.58±0.10**

number of classes for which the highest classification accuracy is obtained is reduced, although ASPS is still the optimal BS method other than the proposed method. MGSR obtained the best classification accuracy in classes 1 and 14, both at 100%, but its OA, AA, and Kappa coefficients are all the worst among the compared methods. Although CS-MF does

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TABLE X
CLASSIFICATION ACCURACY OF PROPOSED HCS-MF AND OTHER STATE-OF-THE-ART ALGORITHMS ON SALINAS BY SVM

Class	ASPS	MVPCA	RDGSR	NGNMF	FNGBS	MGSR	RMGF	CS-MF	HCS-MF
1	99.94±0.07	99.84±0.13	99.86±0.17	99.85±0.14	99.95±0.05	99.22±0.55	99.81±0.14	99.84±0.17	**100.0±0.00**
2	99.92±0.09	99.67±0.18	99.83±0.07	99.83±0.12	99.88±0.09	99.56±0.12	99.84±0.13	99.89±0.10	**100.0±0.00**
3	98.76±0.77	94.74±0.62	97.59±0.36	**99.37±0.61**	99.05±0.49	96.20±0.73	98.78±0.59	98.51±0.57	99.13±0.34
4	98.71±0.35	**98.84±0.25**	98.81±0.25	98.70±0.50	98.69±0.50	98.50±0.42	98.62±0.57	98.37±0.58	98.54±0.75
5	99.54±0.20	98.60±0.52	99.19±0.27	99.44±0.23	99.41±0.25	98.61±0.38	99.56±0.19	99.40±0.24	**99.59±0.18**
6	99.92±0.05	99.49±0.25	99.94±0.04	99.97±0.02	99.95±0.05	99.05±0.26	99.95±0.04	**99.98±0.02**	99.95±0.05
7	99.91±0.12	99.87±0.10	99.86±0.08	**99.93±0.06**	99.90±0.15	99.49±0.29	99.77±0.20	99.74±0.22	99.86±0.15
8	85.63±0.72	83.05±0.32	85.34±0.76	**86.34±0.18**	86.03±0.60	81.93±0.41	84.22±0.76	84.10±1.36	85.79±0.70
9	99.53±0.12	98.80±0.14	99.53±0.06	99.53±0.10	99.53±0.09	99.50±0.11	99.47±0.13	99.35±0.21	**99.54±0.08**
10	96.21±1.21	89.70±1.61	96.69±0.50	96.71±0.70	96.70±0.71	93.88±0.90	96.82±0.67	95.37±0.96	**98.27±1.03**
11	96.87±0.97	75.47±3.80	95.77±1.56	**98.85±0.63**	98.69±0.49	83.42±1.70	98.47±0.77	98.10±0.78	98.45±1.02
12	**99.47±0.19**	98.49±0.42	99.30±0.45	99.20±0.30	99.36±0.20	96.94±1.05	99.11±0.31	99.32±0.49	99.35±0.35
13	99.61±0.42	99.19±0.43	99.41±0.21	99.22±0.51	**99.96±0.09**	97.48±1.25	99.50±0.42	99.24±0.79	99.78±0.31
14	95.33±1.38	92.52±1.72	96.87±1.16	**97.00±0.96**	95.89±1.91	93.65±2.06	96.35±1.65	94.11±1.81	95.33±1.02
15	73.34±0.97	69.30±1.53	72.40±0.84	74.08±0.79	73.89±0.44	67.34±1.37	72.65±1.08	71.15±1.71	**74.69±0.75**
16	98.96±0.57	97.69±0.86	98.52±0.93	**98.70±0.32**	98.52±0.82	97.48±1.01	98.26±0.90	98.35±0.75	98.40±0.56
OA(%)	91.87±0.23	89.51±0.31	91.59±0.15	92.21±0.23	92.10±0.06	89.30±0.29	91.49±0.16	91.03±0.16	**92.23±0.11**
AA(%)	96.35±0.15	93.45±0.28	96.18±0.15	96.67±0.13	96.59±0.17	93.89±0.21	96.32±0.21	95.93±0.15	**96.67±0.12**
Kappa(×100)	90.75±0.26	88.08±0.35	90.43±0.17	91.14±0.25	91.02±0.07	87.84±0.32	90.32±0.19	89.80±0.17	**91.16±0.12**

TABLE XI
CLASSIFICATION ACCURACY OF PROPOSED HCS-MF AND OTHER STATE-OF-THE-ART ALGORITHMS ON SALINAS BY EPF-SVM

Class	ASPS	MVPCA	RDGSR	NGNMF	FNGBS	MGSR	RMGF	CS-MF	HCS-MF
1	**100.0±0.00**	**100.0±0.00**	99.99±0.02	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
2	**100.0±0.00**	99.94±0.07	99.98±0.05	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
3	99.18±0.59	99.68±0.35	99.71±0.38	99.86±0.17	99.57±0.42	99.60±0.60	**99.84±0.22**	99.36±0.53	99.38±0.67
4	98.20±0.32	98.40±0.35	98.47±0.42	98.04±0.54	98.23±0.40	98.41±0.33	**98.68±0.64**	98.14±0.64	98.45±0.39
5	99.87±0.11	**99.96±0.05**	99.84±0.12	99.85±0.09	99.86±0.25	99.90±0.08	99.94±0.07	99.86±0.12	99.90±0.12
6	**100.0±0.01**	**100.0±0.01**	99.94±0.08	**100.0±0.00**	**100.0±0.01**	99.98±0.03	**100.0±0.01**	**100.0±0.00**	**100.0±0.00**
7	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	99.97±0.05	**100.0±0.00**	99.98±0.05	**100.0±0.00**
8	94.33±1.86	92.00±0.95	93.52±2.36	95.40±1.35	96.03±0.41	92.50±1.72	90.71±2.68	88.09±1.53	**97.11±0.72**
9	99.70±0.07	99.47±0.17	99.67±0.08	99.65±0.05	99.56±0.13	99.65±0.08	**99.71±0.08**	99.60±0.09	99.66±0.11
10	97.89±1.08	96.47±1.30	98.20±0.70	97.63±0.25	97.44±0.98	**98.71±1.20**	98.55±0.78	97.87±0.91	98.22±0.97
11	**100.0±0.00**	99.98±0.06	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	99.52±0.62	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
12	**100.0±0.00**	99.87±0.09	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	99.98±0.05	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
13	**100.0±0.00**	99.93±0.11	99.98±0.07	99.87±0.11	99.98±0.07	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
14	99.41±0.36	97.67±0.88	99.22±0.52	99.55±0.27	99.55±0.34	99.39±0.20	**99.87±0.20**	99.18±0.71	99.61±0.39
15	90.35±2.55	87.04±0.94	86.63±1.13	89.44±1.58	89.66±0.57	88.92±1.79	87.47±1.29	86.92±0.96	**90.67±1.93**
16	99.92±0.15	99.95±0.10	99.88±0.17	**100.0±0.00**	99.99±0.03	99.74±0.41	99.96±0.14	99.88±0.19	99.92±0.15
OA(%)	96.93±0.79	95.76±0.29	96.16±0.68	97.03±0.55	97.17±0.14	96.33±0.65	95.71±0.81	94.91±0.39	**97.63±0.33**
AA(%)	98.68±0.27	98.15±0.14	98.44±0.23	98.71±0.15	98.74±0.10	98.52±0.25	98.42±0.27	98.06±0.13	**98.93±0.13**
Kappa(×100)	96.51±0.91	95.17±0.34	95.63±0.78	96.62±0.62	96.78±0.16	95.82±0.74	95.12±0.93	94.20±0.45	**97.30±0.37**

not have the highest classification accuracy in each class, its OA, AA, and Kappa coefficients are still higher than those of NGNMF, RDGSR, and MGSR, which demonstrates the reasonableness and superiority of the MF-based BS model proposed in this article. When using EPF-SVM as a classifier, all BS methods except MVPCA obtained more than 98% classification accuracy, as shown in Table V. The proposed HCS-MF method achieves the highest classification accuracy in nine out of 14 classes, six of which reach 100% classification accuracy. The ASPS and FNGBS methods obtained classification results just below the proposed method, with OA of 98.84% and 98.85%, respectively. Comprehensively analyzing the above experimental results, the proposed HCS-MF method is superior to the other methods in terms of classification results on different classifiers. Compared with CS-MF, the effectiveness of the proposed heterogeneous search strategy is demonstrated.

Fig. 7 shows a classification accuracy map of multiple comparison methods with increasing number of bands on random forest, SVM and EPF-SVM classifiers. For the Botswana dataset, the proposed algorithm achieves satisfactory classification results on random forest and SVM classifiers, as shown in Fig. 7(a) and (b). The classification accuracy of all the methods increase with the number of selected bands. Unlike the above classifier results, for the EPF-SVM classifier, all methods are very close to each other and the proposed HCS-MF algorithm leads only by a small margin as shown in Fig. 7(c).

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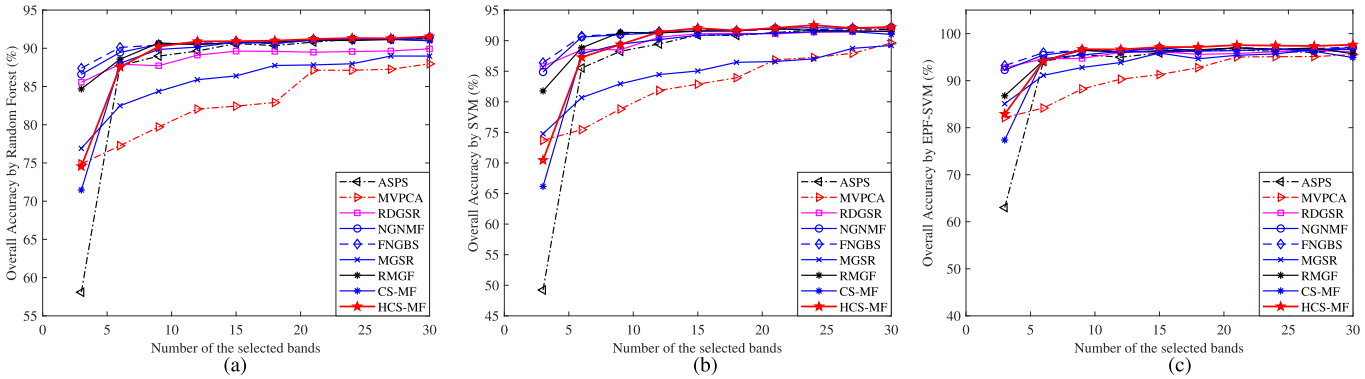


Fig. 9. Results by selecting different number of bands on Salinas. (a) Random forest. (b) SVM. (c) EPF-SVM.

TABLE XII
CLASSIFICATION RESULTS FOR DIFFERENT VERSIONS OF THE CS ALGORITHM

Dataset	Algorithm	Random Forest			SVM			EPF-SVM		
		OA(%)	AA(%)	Kappa(×100)	OA(%)	AA(%)	Kappa(×100)	OA(%)	AA(%)	Kappa(×100)
Botswana	CS-MF	83.36±0.81	84.28±1.02	81.98±0.88	93.30±0.25	93.82±0.33	92.75±0.27	98.19±0.19	98.32±0.21	98.04±0.21
	CS-HL	85.22±0.09	85.92±0.38	83.99±0.10	94.28±0.08	94.80±0.20	93.81±0.09	98.97±0.04	98.97±0.09	98.88±0.04
	CS-HP	84.95±0.11	85.99±0.32	83.70±0.12	94.09±0.05	94.63±0.19	93.60±0.05	98.79±0.11	98.84±0.14	98.69±0.12
	HCS-MF	**86.47±0.46**	**87.31±0.53**	**85.35±0.50**	**95.16±0.24**	**95.62±0.25**	**94.76±0.26**	**99.46±0.14**	**99.46±0.15**	**99.41±0.15**
KSC	CS-MF	84.24±0.27	77.73±0.76	82.43±0.31	93.77±0.23	90.93±0.40	93.06±0.25	99.02±0.12	98.74±0.13	98.91±0.14
	CS-HL	85.43±0.08	79.33±0.64	83.77±0.08	94.65±0.04	92.10±0.34	94.04±0.05	99.50±0.06	99.32±0.10	99.44±0.07
	CS-HP	85.19±0.10	78.95±0.59	83.49±0.11	94.47±0.07	91.61±0.42	93.84±0.08	99.35±0.05	99.14±0.07	99.28±0.06
	HCS-MF	**86.39±0.38**	**80.42±0.61**	**84.83±0.41**	**95.35±0.37**	**93.09±0.73**	**94.82±0.41**	**99.79±0.05**	**99.73±0.09**	**99.77±0.06**
Salinas	CS-MF	90.98±0.04	94.92±0.16	89.95±0.04	91.03±0.16	95.93±0.15	89.80±0.17	94.91±0.39	98.06±0.13	94.20±0.45
	CS-HL	91.28±0.01	95.09±0.10	90.28±0.02	91.68±0.05	96.37±0.17	90.54±0.06	96.79±0.12	98.66±0.08	96.35±0.14
	CS-HP	91.22±0.02	95.12±0.14	90.22±0.02	91.54±0.06	96.27±0.11	90.38±0.06	96.22±0.15	98.51±0.10	95.69±0.17
	HCS-MF	**91.55±0.09**	**95.26±0.07**	**90.58±0.10**	**92.06±0.11**	**96.60±0.12**	**90.97±0.12**	**97.63±0.33**	**98.93±0.13**	**97.30±0.37**

TABLE XIII
CLASSIFICATION RESULTS UNDER MULTISCALE CROSS-BRANCH RESPONSE AND SECOND-ORDER CHANNEL ATTENTION

Dataset		OA(%)	AA(%)	Kappa(×100)
Botswana	13	93.43±1.24	94.39±1.01	92.89±0.98
	30	**99.52±0.57**	**99.59±0.41**	**99.48±0.13**
	All	99.21±0.49	99.33±0.19	99.15±0.84
KSC	14	98.64±1.01	97.66±0.84	98.49±0.17
	30	**99.85±1.05**	**99.76±0.98**	**99.83±1.57**
	All	99.45±2.10	98.74±3.01	99.38±2.93
Salinas	18	99.66±1.08	99.53±2.45	99.62±1.76
	30	**99.94±1.64**	99.85±1.84	**99.83±1.74**
	All	99.79±1.41	**99.89±2.10**	99.77±1.98

E. Kennedy Space Center

For the KSC dataset, the numerical classification results of the proposed HCS-MF algorithm and the comparison methods are shown in Tables VI–VIII, including the OA, AA, and Kappa coefficients, as well as the classification accuracy for each class. Fig. 8 shows the classification accuracy of the HCS-MF method and the other methods on Random Forest, SVM and EPF-SVM classifiers with increasing number of the selected bands.

The classification results of the HCS-MF method with random forest as a classifier and the other methods are shown in Table VI. The results show that the proposed HCS-MF method has the highest classification accuracy in seven out of 13 classes, followed by FNGBS. The OA obtained by the proposed method and FNGBS method is ahead of other methods with 86.39% and 86.03%, respectively. The experimental results with SVM as the classifier are shown in Table VII, where the proposed HCS-MF algorithm obtains the highest classification accuracy in eight out of 13 classes. The proposed method, CS-MF, NGNMF, FNGBS, and RMGF methods all obtained 100% classification accuracy in class 13, in addition, RMGF obtained 100% classification accuracy in class 10. As shown in Table VIII, all the BS methods can achieve 100% classification accuracy in some of the classes using EPF-SVM as the classifier, where the proposed HCS-MF method has the highest classification accuracy in 11 out of 13 classes. The proposed algorithm obtains the optimal classification result with OA of 99.79%, followed by the FNGBS method, which has an OA of 99.65%.

Fig. 8 shows a classification accuracy map of multiple comparison methods with increasing number of bands on random forest, SVM and EPF-SVM classifiers. For the KSC dataset, the HCS-MF method has a better performance when the number of selected bands is more than 10. From the classification results of the three classifiers, it can be seen that MVPCA, MGSR, and RDGSR provide the worst classification results, almost 10% below the classification accuracy of the other comparison algorithms. Only the classification performance of the RDGSR method approaches the other methods

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TABLE XIV
EXECUTION TIME OF DIFFERENT METHODS BY SELECTING A CERTAIN NUMBER OF BANDS ON THREE DATASETS (S)

Dataset	No.	ASPS	MVPCA	RDGSR	NGNMF	FNGBS	MGSR	RMGF	CS-MF	HCS-MF
Botswana	13 bands	0.5604	0.7483	3.1550	1.6184	0.4386	2.6646	27.0823	0.8036	0.8354
	30 bands	0.5826	0.7474	3.3565	1.5788	0.4319	2.7705	27.2562	1.0951	1.1464
KSC	14 bands	0.8052	0.7064	6.0548	1.6978	0.4297	3.2930	30.6987	0.8925	0.9341
	30 bands	0.9349	0.7062	5.6639	1.7700	0.4269	3.2454	30.1626	1.1732	1.2461
Salinas	18 bands	0.4612	0.1220	425.3993	0.7212	0.1897	218.8028	25.9912	1.0345	1.0895
	30 bands	0.5013	0.1237	426.6825	0.7503	0.1904	224.2683	26.0735	1.2548	1.2894

when the number of selected bands is 20 on the EPF-SVM classifier.

F. Salinas

The classification results of the proposed HCS-MF method for the Salinas dataset and the comparison methods on the random forest, SVM, and EPF-SVM classifiers are shown in Tables IX–XI, including OA, AA, Kappa coefficients, and classification accuracy of each class. Fig. 9 shows the classification results of the proposed HCS-MF method and the other methods on the above classifiers for subsets consisting of different numbers of selected bands.

As the experimental results in Table IX show, the classification accuracy of the proposed HCS-MF method with random forest as the classifier is ahead of the other methods by a slight margin, where the method with the highest performance among the other methods is FNGBS. The proposed method has the highest classification accuracy in six out of 16 classes. For classification accuracy for each class when SVM is the classifier, both the proposed HCS-MF method and NGNMF have the highest classification accuracy for six out of 16 classes, but the former obtains 100% classification accuracy on classes 1 and 2. Simultaneously the proposed algorithm and the NGNMF method are also ahead of the other compared methods in terms of overall classification accuracy, with OA of 92.23% and 92.21%, respectively. From Table XI, the proposed method achieves a classification accuracy OA of 97.63% with EPF-SVM as the classifier, followed closely by FNGBS with 97.17%, and the worst one is the CS-MF. The HCS-MF method can obtain the highest classification accuracy in nine out of 16 classes and 100% classification accuracy in seven classes. For class 16, only the NGNMF method obtains 100% classification accuracy.

The classification accuracy of the proposed HCS-MF method and the other methods on different classifiers with different number of selected bands is shown in Fig. 9. For the Salinas dataset, the results on the three classifiers differ from the two datasets mentioned above, with all methods achieving approximately the same OA, except the ASPS method performs poorly when the number of bands is small. As can be seen in Fig. 9, the proposed HCS-MF method can still classify the samples better, but the difference is not obvious.

G. Ablation Experiment

To demonstrate the effectiveness and superiority of the heterogeneous search strategy proposed in this article, CS with heterogeneous Lévy flight (CS-HL), CS with heterogeneous preference random wandering (CS-HP), and the proposed HCS-MF algorithm are compared, respectively, and the experimental results are shown in Table XII. As can be seen from Table XII, the OA of CS-HL and CS-HP algorithms on all three datasets is improved compared to CS-MF, which indicates that embedding heterogeneous searches in the global exploration and local exploitation phases of the CS algorithm, respectively, is conducive to the selection of higher quality band subsets. Whereas, compared to the HCS-MF algorithm, is demonstrated that embedding heterogeneous search in each stage of the CS algorithm can achieve better classification performance.

To study the performance of the proposed HCS-MF method under multiscale cross-branch, experiments on the recommended number of selected bands, 30 selected bands and all bands per dataset are performed under the HSI classification method based on multiscale cross-branching response and second-order channel attention (MCRSCA) [49] respectively, 10% of the training samples in the experiment are randomly selected, and the experimental results are shown in Table XIII. For Botswana and KSC, the 30 selected bands provided by the proposed HCS-MF method are able to obtain the high classification accuracy under MCRSCA, which is higher than all-bands. For Salinas, the classification performance of the 30 selected bands (AA 99.85%) is lower than the full-band classification performance (AA 99.89%) only in AA. The experimental results demonstrate that the bands selected by the proposed HCS-MF method have lower correlation, more representative and higher signal-to-noise ratio.

H. Computational Time Comparison

In this section, to highlight the superiority of the execution time of the proposed method, all the BS methods are performed in MATLAB 2020a using a PC workstation (Intel Core i9-11900K CPU processor and 32 GB RAM). The execution times of all the BS methods for selecting different numbers of bands in the three datasets are shown in Table XIV. The experimental results show that the RDGSR, RMGF, and MGSR methods have very high execution time, while the other methods have lower processing times. For the Salinas dataset, the execution time for both RDGSR and MGSR methods exceeds 200 s, which is excessively time-costly. Compared with the CS-MF method, the execution time of the proposed HCS-MF method exceeds only about 0.05 s. Obviously, it is

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significant and essential to improve the classification performance within an acceptable time. Compared with NGNMF and FNGBS methods, the execution time of the proposed method is lower than that of NGNMF and higher than that of FNGBS method, but the classification performance achieves satisfactory results. Overall, the proposed method not only performs faster, but also outperforms the other state-of-the-art methods in classification on all three datasets.

V. Conclusion

In this article, an unsupervised BS method based on HCS algorithm with MF is proposed for HSI classification. First, a heterogeneous search strategy is adopted to improve the exploration and exploitation capability of CS algorithm. Then, a matching filter-based BS model is designed to select a subset of bands with high signal-to-noise ratio. Meanwhile, a mapping method based on neighborhood grouping strategy is designed to reduce the similarity among the selected bands. Subsequently, an automatic recommendation strategy for the number of selected bands based on SSD is proposed. Finally, extensive experimental results demonstrate the effectiveness of the proposed HCS-MF algorithm, and the superiority of the proposed algorithm in comparison with the state-of-the-art BS methods such as FNGBS, MGSR, and RMGF.

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Meng Wu received the B.S. degree from Zunyi Normal University, Zunyi, China, in 2020. He is currently pursuing the M.S. degree with the Hunan Institute of Science and Technology, Yueyang, China.

His research interests include evolutionary computing, and hyperspectral image classification, and band selection.



Xianfeng Ou received the M.S. degree in communication and information system from Xinjiang University, Urumchi,, China, in 2009, and the Ph.D. degree in communication and information system from Sichuan University, Chengdu, China, in 2015.

He is currently an Associate Professor with the School of Information Science and Technology, Hunan Institute of Science and Technology, Yueyang, China. His research interests include image and video coding processing technologies, object detection, hyperspectral image classification, and change detection.



Youli Lu received the M.S. degree in computer application technology from the Changsha University of Science and Technology, Changsha, Hunan, China, in 2010.

She is currently an Associate Professor with the School of Information Science and Technology, Hunan Institute of Science and Technology, Yueyang, China. Her research interests include big data mining and hyperspectral image classification.



Wujing Li received the B.S. degree in software engineering and the Ph.D. degree in computer science and technology from Sichuan University, Chengdu, China, in 2007 and 2012, respectively.

He is currently a Lecturer with the School of Information Science and Technology, Hunan Institute of Science and Technology, Yueyang, China. His research interests include image denoising, image restoration, image enhancement, and hyperspectral image processing.



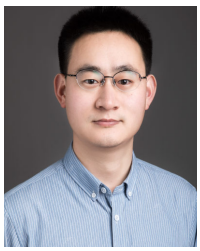
Dan Yu received the M.S. degree in optical engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2006.

He is currently a Lecturer with the School of Information Science and Technology, Hunan Institute of Science and Technology, Yueyang, China. His research interests include image processing technologies, hyperspectral image classification, and change detection.



Zhihao Liu is currently pursuing the B.S. degree with the Hunan Institute of Science and Technology, Yueyang, China.

His main research interests include hyperspectral image processing.



Chengtao Ji (Member, IEEE) received the M.S. degree in communication and information system from Sichuan University, Chengdu, China, in 2014, and the Ph.D. degree in computer science from the University of Groningen, Groningen, The Netherlands, in 2018.

He is currently an Assistant Professor with the Computing Department, Xi'an Jiaotong-Liverpool University, Suzhou, China. His research interests include machine learning, hyperspectral image processing, big data mining, and visualization.

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